Unit-3

(Ethereum)

Syllabus:- Introduction, History of Ethereum, Features of Ethereum, Ethereum Accounts, Applications of Ethereum, DAO, EVM-Ethereum Virtual Machine, Understanding Gas In Ethereum, Working Process of Ethereum, Consensus Mechanisms, Smart Contracts, Metamask Setups

What is Ethereum?

- Ethereum is a decentralized open source blockchain system that features its own cryptocurrency ether (ETH).
- Ethereum is not a cryptocurrency. Ethereum has its own cryptocurrency, Ether, which is used to pay for certain activities on the Ethereum network.
- Ethereum is an open source blockchain system and provides a platform. This platform can be used for various DAPPS which can be deployed by using smart contracts. DAPPS or decentralized apps are DeFi's and NFT's.
- Ethereum is defined as a blockchain-based computing platform that enables its developers to build organizations, holding assets, deploy decentralized applications ,transacting and communicating without being controlled by a central authority.
- Ether is the cryptocurrency for Ethereum, which is used to build decentralized applications, smart contracts, and make standard peer-to-peer payments.
- Ether acts as fuel for the Ethereum Network. It tracks and facilitates all the transactions in the network.

KEY TAKEAWAYS

- **Ether:** This is Ethereum's cryptocurrency.
- **Smart contracts:** Ethereum allows the development and deployment of these types of contracts.
- **Ethereum Virtual Machine (EVM):** Ethereum provides the underlying technology—the architecture and the software—that understands smart contracts and allows you to interact with it.
- **Decentralized applications (Dapps):** A decentralized application is called a Dapp (also spelled DAPP, App, or DApp) for short. Ethereum allows you to create consolidated applications, called decentralized applications.
- **Decentralized autonomous organizations (DAOs):** Ethereum allows you to create these for democratic decision-making.

How is Ethereum different than Bitcoin?

While there are many similarities between Ethereum and Bitcoin, there are also significant differences. Here are a few:

1. Bitcoin trades in cryptocurrency, while Ethereum offers several methods of exchange, including cryptocurrency (Ethereum's is called Ether),
2. Based on different security protocols: Ethereum uses a "proof of stake" system as opposed

the "proof of work" system used by Bitcoin.

3. Bitcoin allows only public (permissionless or censor-proof) transactions to take place; Ethereum allows both permissioned and permissionless transactions.
4. The average block time for Ethereum is significantly less than Bitcoin's: 12 seconds versus 10 minutes.
5. It is estimated that by 2021 only half of the Ether coins will be mined (a supply of more than 90 million tokens), but the majority of Bitcoins already have been mined (its supply is capped at 21 million).
6. For Bitcoin, the computers (called miners) running the platform and verifying the transactions receive rewards. Basically, the first computer that solves each new block gets Bitcoins (or a fraction of one) as a reward. Ethereum does not offer block rewards and instead allows miners to take a transaction fee.

Ethereum Benefits

- **Large, existing network.** The benefits of Ethereum are a tried-and-true network that has been tested through years of operation and billions of value trading hands. It has a large and committed global community and the largest ecosystem in blockchain and cryptocurrency.
- **Wide range of functions.** Besides being used as a digital currency, Ethereum can also process other financial transactions, execute smart contracts and store data for third-party applications.
- **Constant innovation.** A large community of Ethereum developers is constantly looking for new ways to improve the network and develop new applications. "Because of Ethereum's popularity, it tends to be the preferred blockchain network for new and exciting (and sometimes risky) decentralized applications," Avital says.
- **Avoids intermediaries.** Ethereum's decentralized network promises to let users leave behind third-party intermediaries, like lawyers who write and interpret contracts, banks that are intermediaries in financial transactions or third-party web hosting services.

What Is the Main Difference in Application Between Bitcoin and Ethereum?

Bitcoin is primarily designed to be an alternative to traditional currencies and hence a medium of exchange and store of value. Ethereum is a programmable blockchain that finds application in numerous areas, including DeFi, smart contracts, and NFTs.

How Many BTC and ETH are Currently in Circulation?

As of Oct. 26, 2023, there were 19.52 million BTC and 120.26 million ETH in circulation.

What Are Some Similarities Between Bitcoin and Ether?

Ether and bitcoin are alike in many ways. Each digital currency is traded on online exchanges and stored in cryptocurrency wallets. Both are decentralized, meaning they are not issued or regulated by a central bank or other authority, and both use blockchain technology.

How does Ethereum work?

- Like Bitcoin, the Ethereum network exists on thousands of computers worldwide, thanks to users participating as "nodes," rather than a centralized server. This makes the network decentralized and highly immune to attacks, and essentially unable to go down as a result. If one computer goes down, it doesn't matter because thousands of others are holding the network up.

- Ethereum is essentially a single decentralized system that runs a computer called the Ethereum Virtual Machine (EVM). Each node holds a copy of that computer, meaning that any interactions must be verified so everyone can update their copy.

- Network interactions are otherwise considered "transactions" and are stored within blocks on the Ethereum blockchain. Miners validate these blocks before committing them to the network and acting as transaction history or a digital ledger. Mining to verify transactions is known as a proof-of-work (PoW) consensus method. Each block has a unique 64-digit code identifying it. Miners commit their computer power to find that code, proving that it's unique. Their computer power is "proof" of that work, and miners are rewarded in ETH for their efforts.

- Also, like Bitcoin, all Ethereum transactions are entirely public. Miners broadcast completed blocks to the rest of the network, confirming the change and adding the blocks to everyone's copy of the ledger. Confirmed blocks cannot be tampered with, serving as a perfect history of all network transactions.

- But, if miners are paid for their work, where does that ETH come from? Each transaction comes with a fee, called "gas," which is paid by the user initiating the said transaction. That fee is paid to the miner who validates the transaction, incentivizing future mining and ensuring network security. Gas essentially serves as a limit, restricting the number of actions a user can make per transaction. It's also in place to prevent network spam.

- Because ETH is more of a utility token than a token of value, its supply is infinite. Ether consistently enters circulation in the form of miner rewards, and it will with staking rewards as well once the network moves to proof-of-stake (PoS). In theory, Ether will always be in demand, meaning inflation should never devalue the asset beyond use.

- Unfortunately for many, Ethereum gas fees can run quite high based on network activity. This is because a block can only hold so much gas which varies based on transaction types and amounts. As a result, miners will choose transactions with the highest gas fees, meaning users are competing to validate transactions first. This competition pushes fees higher and higher, congesting the network during busy times.

- Network congestion is a significant problem, though it's being addressed in Ethereum 2.0 — a complete overhaul that will be discussed as a separate section.

- Interacting with Ethereum requires cryptocurrency, which is stored in a wallet. That wallet connects to DApps, acting as a passport for the Ethereum ecosystem. From there, anyone can purchase items, play games, lend money and do all sorts of activities just as they do on the traditional internet. Only, the traditional web is free to users, as they're giving away personal information. Centralized entities running websites then sell that data to make money.

- Cryptocurrency takes the place of data here, meaning users are free to browse and interact anonymously. This also means DApp use is nondiscriminatory. For example, no lending or banking DApp can reject someone based on their race or financial status. An intermediary can't block what they consider a "suspicious transaction." Users control what they do and how they do it, which is why many consider Ethereum to be Web 3.0 — the future of web interaction.

Differentiate Bitcoin vs Ethereum

Basis	Bitcoin	Ethereum
Definition	Bitcoin (abbreviation: BTC; sign: ₿) is a decentralized digital currency that can be transferred on the peer-to-peer bitcoin network.	Ethereum is a decentralized global software platform powered by blockchain technology. It is most commonly known for its native cryptocurrency, ether (ETH).
History	The word bitcoin was defined in a white paper published on 31 October 2008. The currency began use in 2009.	Ethereum was conceived in 2013 by programmer Vitalik Buterin, and then went live on 30 July 2015.
Purpose	The purpose of bitcoin was to replace national currencies during the financial crisis of 2008.	The purpose of Ethereum was to utilize blockchain technology for maintaining a decentralized payment network and storing computer code.
Smart Contracts	Although bitcoin do have smart contracts, they are not as flexible or complete as Ethereum smart contracts. Smart contracts in Bitcoin does not have all the functionality that a programming language would give them.	Ethereum allows us to create smart contracts. Smart contracts are computer codes that is stored on a blockchain and executed when the predetermined terms and conditions are met.
Smart Contract Programming Language	Smart contracts on Bitcoin are written in programming languages like Script, Clarity.	Smart contracts on Ethereum are written in programming languages like Solidity, Vyper, etc.
Transactions	Generally, bitcoin transactions are only for keeping notes.	Ethereum transactions may contain some executable code.
Hash Algorithm	Bitcoin runs on the **SHA-256** hash algorithm.	Ethereum runs on the **Keccak-256** hash algorithm.

Basis	Bitcoin	Ethereum
Consensus Mechanism	The Proof-of-Work (PoW) is the consensus mechanism used by the Bitcoin network.	The Proof-of-Stake is the consensus mechanism used by Ethereum.
Block Time	The block time of bitcoin is 10 minutes.	The block time of Ethereum is 14 to 15 seconds.
Block Limit	The bitcoin blockchain has a block limit of 1 MB.	The Ethereum blockchain does not have a block limit.
Popularity	Bitcoin is the most popular digital currency in the market to date.	Ether, native currency of Ethereum is the second-largest cryptocurrency after bitcoin to date.
Supply	The majority of Bitcoins already have been mined (its supply is capped at 21 million).	It is estimated that by 2021 only half of the Ether coins will be mined (a supply of more than 90 million tokens)

History of Ethereum

- A non-profit organisation called the Ethereum Foundation was formed on July 6th, 2014 and registered in Zug, Switzerland. This organisation governs the Ethereum project.
- Vitalik Buterin is one of the Ethereum's co-founder,
- Buterin was introduced and charmed by blockchain technology when he got involved in Bitcoin as a 17-year-old programmer in 2011 and co-founded Bitcoin Magazine

The four phases of Ethereum

Frontier (July 30th, 2015):

Olympic, a test version of Frontier, was released on May 9th, 2015. The purpose was to test the project and reward those who found issues.

Homestead (March 14th, 2016):

The platform enhanced with Ethereum Improvement Proposals (EIP). The upgrade helped the future updates and improved the speed of transactions.

Metropolis (October 16th, 2017):

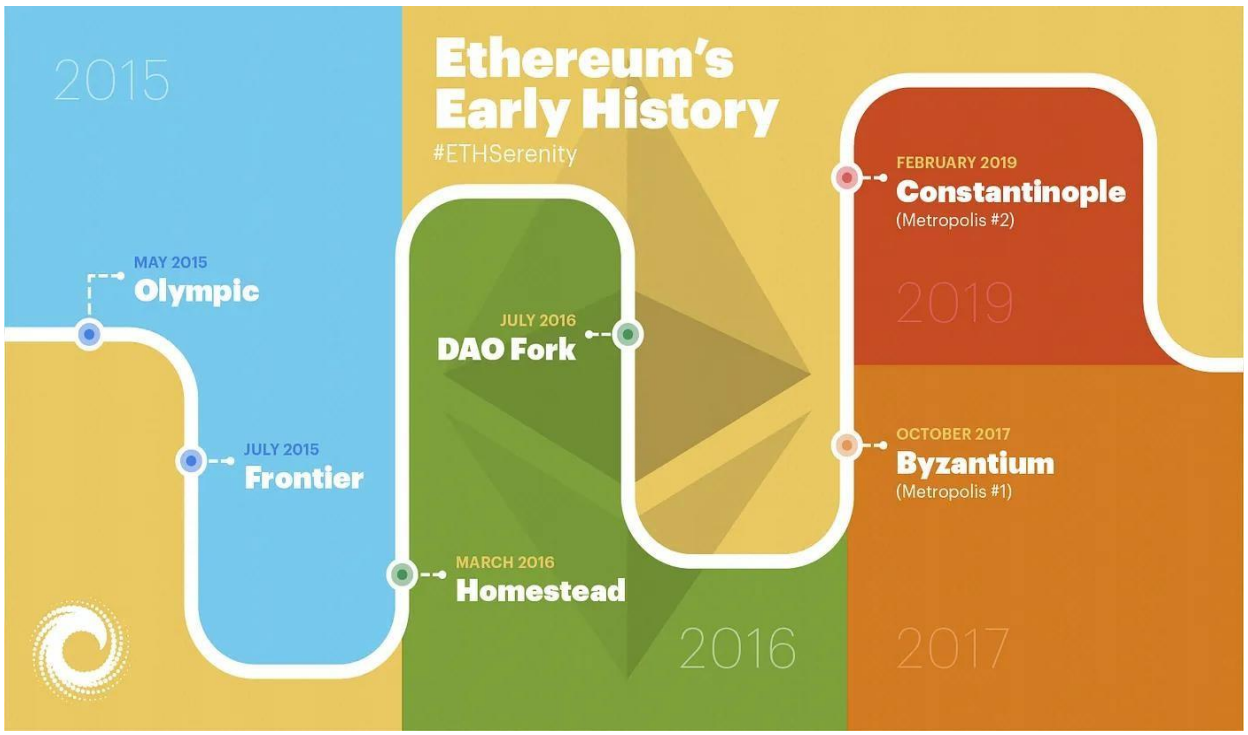
This phase includes two upgrades.

1. **Byzantium (October 2017)**
2. **Constantinople (Scheduled for early 2019)**

The platform improved furthermore in terms of scalability, transaction processing, and security through the Byzantium upgrade

- **Serenity (Ethereum 2.0):**

In this phase, the Ethereum project will move from proof of work (PoW) to proof of stake (PoS), which uses the Casper consensus algorithm. PoS has significant benefits over PoW in terms of security and being resistant to attacks, and also boosts transaction processing times.



Yearwise Evolvement in Ethereum:

- **2013:** Ethereum was first described in Vitalik Buterin's white paper in 2013 with the goal of developing decentralized applications.
- **2014:** In 2014, EVM was specified in a paper by Gavin Wood, and the formal development of the software also began.
- **2015:** In 2015, Ethereum created its genesis block marking the official launch of the platform.
- **2018:** In 2018, Ethereum took second place in Bitcoin in terms of market capitalization.
- **2021:** In 2021, a major network upgrade named London included Ethereum improvement proposal 1559 and introduced a mechanism for reducing transaction fee volatility.
- **2022:** In 2022, Ethereum has shifted from PoW(Proof-of-Work) to PoS(Proof-of-State) consensus mechanism, which is also known as Ethereum Merge. It has reduced Ethereum's energy consumption by ~ 99.95%.

What are Smart Contracts?

In 1994, **Nick Szabo** (inventor of BitGold and a computer scientist) proposed the concept of Smart Contracts. He defined a smart contract as computerized transaction protocols that execute the rules defined in the contract. Years later, **Vitalik Buterin** (co-founder of Ethereum) reinstated the concept and created EVM to execute smart contracts.

Ethereum is a public blockchain and a cryptocurrency (ETH). The Ethereum network has multiple applications, such as **Smart Contracts**.

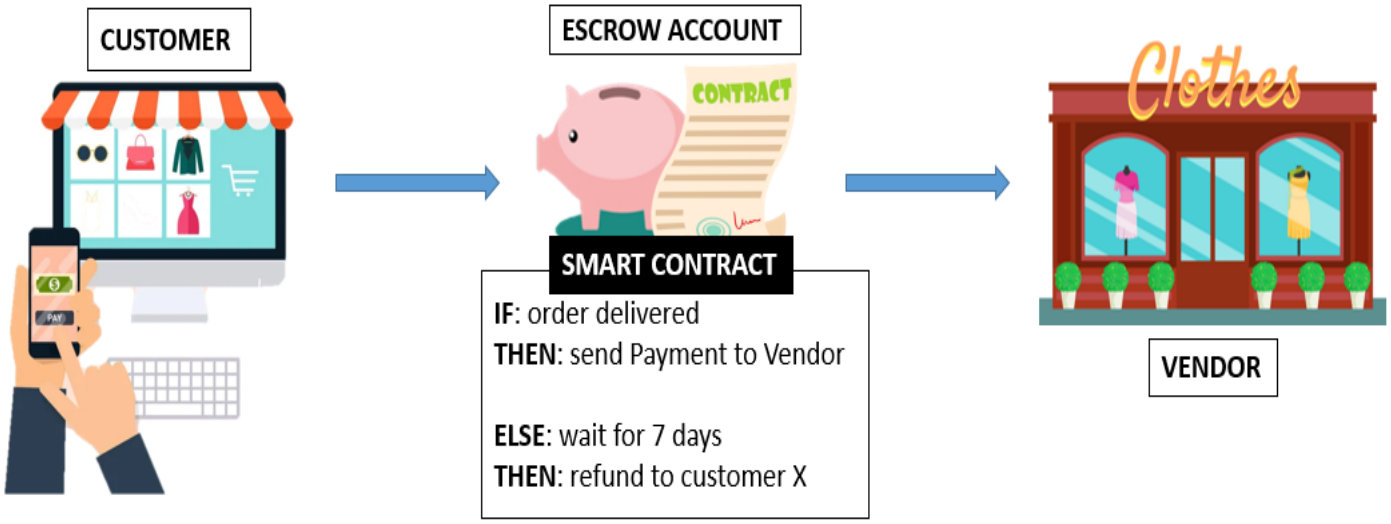
Smart contracts allow different parties to agree on a few conditions that later get coded into a self-executing program. Hence, it's **a software program that automatically executes when the coded conditions are met**. A smart contract eliminates dependency and trust on the third party, saving constant attention, time and cost.

*A **smart contract** includes an automatically-executable program of the business logic or requirement mentioned. It follows if-then statements. If a particular condition is true, then do a specific operation. The smart contract programs are executed by Ethereum Virtual Machine (EVM). Once the program gets executed, the action will be taken automatically as defined in the contract.*

Let's take a simple example to understand the above.

For instance, what if there's a smart contract established between customers and vendors on online shopping?

- On a prepaid order, the payment by the customer goes to an escrow account.
- The payment will not be transferred to the vendor until the customer gets their order.
- Once the customer receives the order, the payment automatically transacts to the vendor.
- If the order does not deliver, the payment goes back to the customer after a specific number of days.



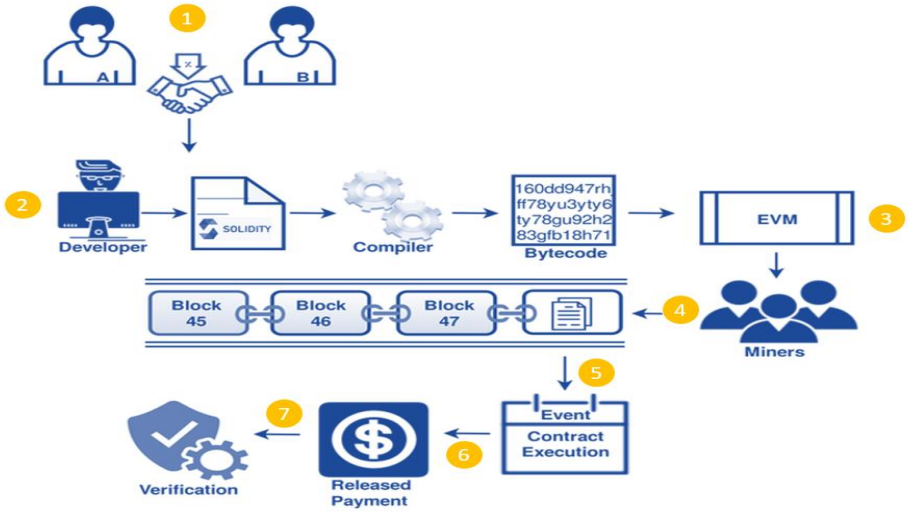
Features of Smart contract:

- Self-executable code
- Deployed on Blockchain networks like Ethereum
- Follows IF-THEN conditions for contract
- Irreversible, can't alter later
- Traceable and Transparent
- An Agreement between Buyer and Seller
- Secure and Reliable

How does Smart Contract work?

Following are the steps to run an Ethereum smart contract:

1. Contract deal between buyer (party A) and the seller (party B).
2. A developer writes the smart contract in the solidity programming language.
3. Solidity code (converted into) → Opcodes (compiles into) → Byte code
4. Byte code (low-level machine language) saved by Ethereum Virtual Machine (EVM).
5. Miners convert the smart contract into a new block to add to the Ethereum blockchain network.
6. When the conditions meet, the contract automatically executes. Hence, payment release to the seller (party B).
7. Verification of payment transaction.



There are numerous applications of Smart Contracts, such as a Voting System, Insurance, crowdfunding, identity verification, supply chain, satellite communication, healthcare, etc.

Benefits of Smart contracts

Following are the advantages of Ethereum-based Smart Contracts:

- Smart contracts are automatically executable programs.
- It removes the need for middle parties or intermediaries. Hence, it eliminates the risk of manipulation and cost-cutting.
- No fear of data loss as the contract has multiple copies shared across the blockchain network.
- Impressions of digital signing and cryptography ensure the safety and authenticity of the contract.
- Saves time and effort in producing real business contracts.
- Eliminate manual human errors while filling out forms.

Limitations of Smart contracts

Following are the disadvantages of Ethereum-based Smart Contracts:

- Once the smart contract code is deployed, it's impossible to alter it.
- There are possibilities of loopholes in the contract
- Not a complete elimination of intermediaries as developers requires them to establish rules for smart contact.
- Not legally banded yet. No approval by any jurisdiction.

Smart Contracts Vs. Traditional Contract Systems

In conventional contract systems, you sign an agreement, then you trust and hire a third party for its execution. The problem is that in this type of process, data tampering is possible. With smart contracts, the agreement is coded in a program.

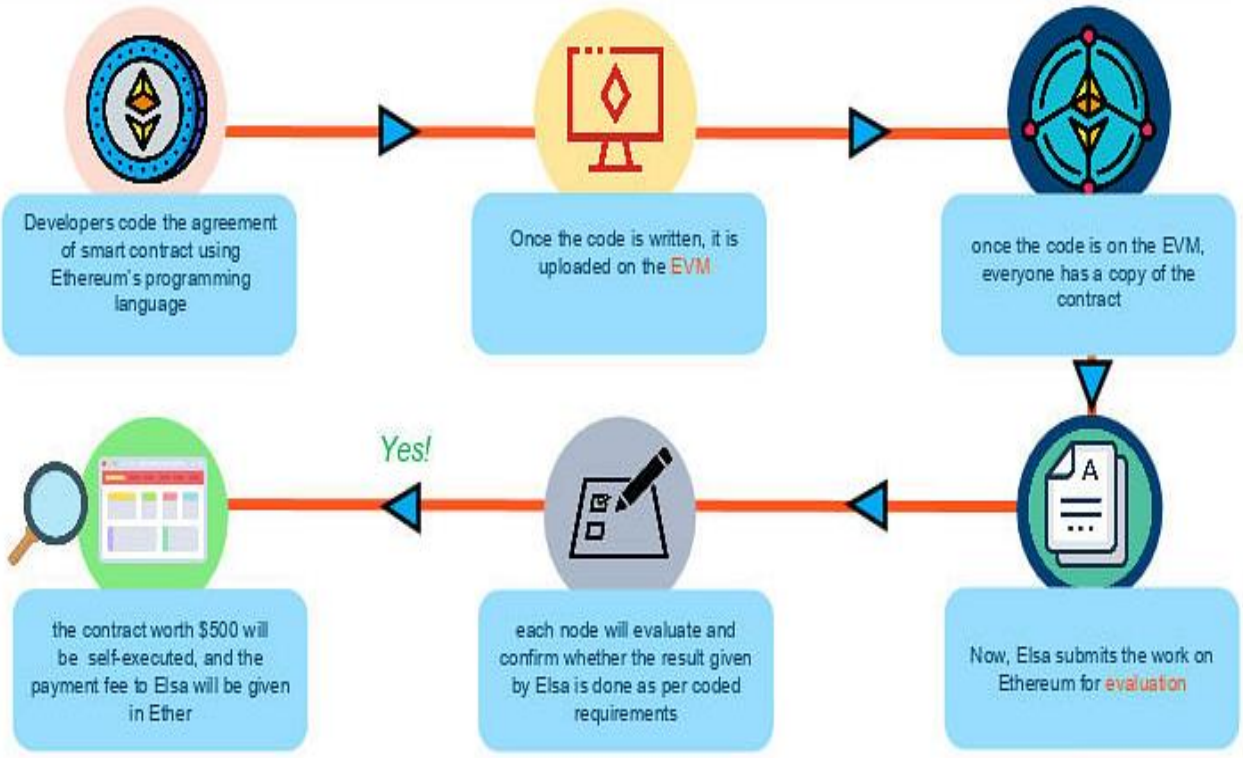
A centralized authority does not verify the result; it is confirmed by the participants on the Ethereum blockchain-based network. Once a contract is executed, the transaction is registered and cannot be altered or tampered, so it removes the risk of any data manipulation or alteration.

Let's take an example in which someone named Zack has given a contract of $500 to someone named Elsa for developing his company's website. The developers code the agreement of the smart contract using Ethereum's programming language.

The smart contract has all the conditions (requirements) for building the website. Once the code is written, it is uploaded and deployed on the Ethereum Virtual Machine (EVM).

EVM is a runtime compiler to execute a smart contract. Once the code is deployed on the EVM, every participant on the network has a copy of the contract. When Elsa submits the work on Ethereum for evaluation, each node on the Ethereum network will evaluate and confirm whether the result given by Elsa has been done as per the coding requirements.

Once the result is approved and verified, the contract worth $500 will be self-executed, and the payment will be paid to Elsa in ether. Zack's account will be automatically debited, and Elsa will be credited with $500 in ether.



What is a dApp?

The browser you're reading this article on interacts with a centralized server to fetch this information for you. Tech giants like Meta (Facebook), Amazon, Google, etc., feeds on customer data. Hence, you're the product for them to generate revenue.

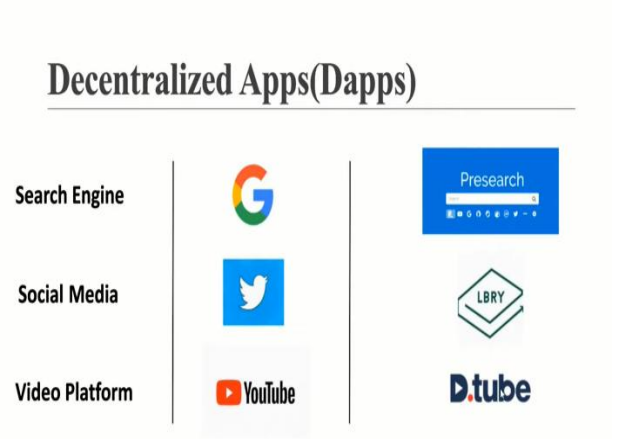
Decentralized applications (dApps) are one of several paths to bring back users' data liberty out of centralized control. Hence, instead of reaching a centralized server, your requests actually contact the Blockchain for information. **dApps are the apps only, without centralized control.**

App = Frontend + Backend → Hosted on Centralized Network Servers.
DApp = Frontend + Backend + Smart Contracts → Hosted on Blockchain.

Smart contracts help to encode the regulations and conditions of the decentralized application. It works along with the Backend of the application. Usually, smart contracts are programmed in solidity language and executed by Ethereum Virtual Machine (EVM).

How does a dApp attain decentralization on Apps?

Decentralized applications are hosted on the blockchain network. It means each node contains the exact same copy of the coded application. Hence, no central server is responsible for changes and maintenance of the application.

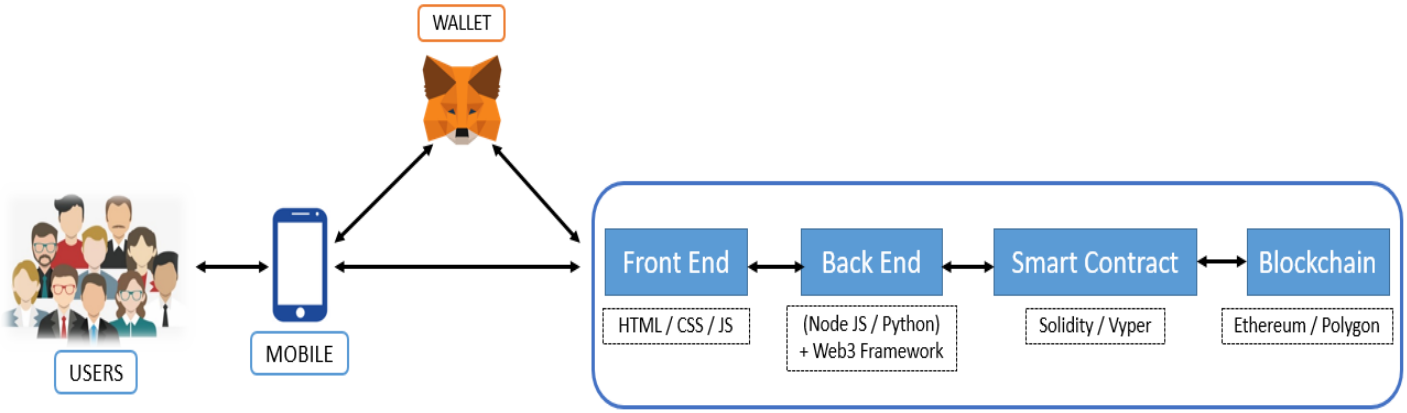


Centralized Apps	Decentralized Apps
Not Trustworthy	Trustworthy
Censorship	No censorship
You pay	They pay
Go down	Can never go down

How Does a dApp Work?

dApps interact with users on mobile or browser like a normal app or website. Users can also connect or log in via wallet to access the application. The dApp hosts on the blockchain network. The source code is available for verification to each node in the network.

The front end of the application is written in HTML, CSS, JS, etc. Backend is written in JS or Python, usually interacting with smart contracts and Blockchain. Moreover, frameworks like NodeJS and Web3 or Django help to create the Backend.



Benefits of dApps

Following are the advantages of Decentralized Applications (dApps):

- **Data integrity and Privacy**
- **Censorship Resistant:** Nobody is authorized to control the application's actions. Hence, equality and **decentralization**.
- **No Downtime**: As the whole network is decentralized, no one or central point of failure.
- **Open-Source** and Permissionless
- Provides **incentives** for valuable reviews and building reputation
- **No third-party trust**

Limitations of dApps

Following are the disadvantages of Decentralized Applications (dApps):

- Decentralization might **affect the User experience and maintenance** as no single entity is responsible.
- Once a smart contract is deployed on the Blockchain, it can't be altered.
- dApp can lead to **network congestion** due to heavy computation.
- Performance overhead
- Scalability issues

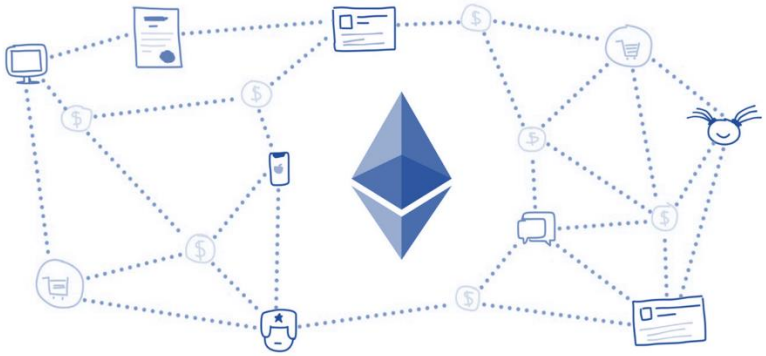
Examples of dApps

Following are a few of the application examples of dApps:

- **Augur:** A decentralized prediction marketplace based on Ethereum
- **Bit Torrent:** A decentralized file-sharing application
- **Golem:** A decentralized marketplace for renting computational power and resources
- **Cryptokitties:** A decentralized game application
- **Minds:** A decentralized social media platform between content creators and consumers

Ethereum Virtual Machine (EVM)

Ethereum is a decentralized public blockchain and a cryptocurrency (ETH). The Ethereum network has multiple applications. One of them is **Smart Contracts**.



the concept of blockchain technology allows for the addition of a new block in the network under consensus algorithms. The smart contracts are also added as a new block in the Ethereum network. Hence, they are irreversible, traceable, and cryptographically secured.

The smart contracts are executed on **Ethereum Virtual Machine (EVM)** by developers building decentralized applications (dApps). These are written in Solidity programming language. However, smart contracts in solidity can't be executed by EVM directly. First, it converts solidity code into low-level machine language Opcodes. Then those opcodes convert into bytecode. Hence, the bytecode of the smart contract gets executed by EVM.

The execution process of the smart contract initiates the Gas amount. **Ethereum Gas** is a transaction fee charge for adding a new block to the network. The gas amount depends on the size of the contract and execution urgency. The longer you wait, the lesser will be the gas fees for you.

This additional amount uses to reward miners and strengthen the global financial blockchain system.

What Is Gas (Ethereum)?

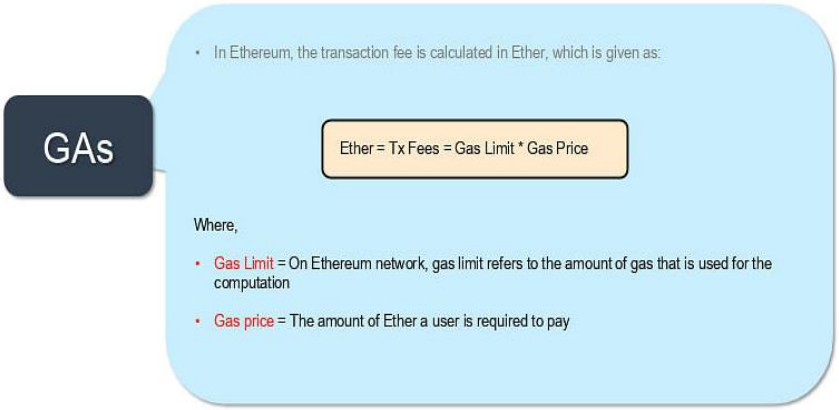
- The concept of gas was introduced to compensate miners for their work done on maintaining and securing the blockchain. Ater the proof of stake algorithm was rolled out in September 2022, gas fees became the reward for staking ETH and participating in validation—the more a user has staked, the more they can earn.
- "Gas limit" is the maximum amount of work you're estimating a validator will do on a particular transaction. A higher gas limit usually means the user believes the transaction will require more work. "Gas price" is the price per unit of work done. So, a transaction cost is the gas limit multiplied by the gas price. Many transactions also include tips, which are added to the gas price (the more you pay, the faster your transaction is completed). The lower a user estimates their gas limit, the lower the priority in the queue they will be.

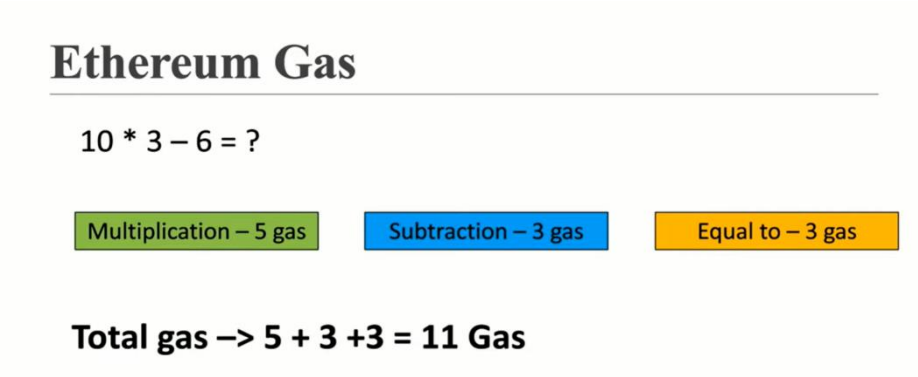
- Gas is the fee required to successfully conduct a transaction or execute a contract on the Ethereum blockchain platform. Fees are priced in tiny fractions of the cryptocurrency ether (ETH)—denominations called gwei (10^{-9} ETH). Gas is used to pay validators for the resources needed to conduct transactions.1
- The exact price of the gas is determined by supply, demand, and network capacity at the time of the transaction.

- Just like we need fuel to run a car, we need gas to run applications on the Ethereum network. To perform any transaction within the Ethereum network, a user must make a payment, in this case paying out ethers, to get a transaction done, and the intermediary monetary value is called gas.

- On the Ethereum network, gas is a unit that measures the computational power required to run a smart contract or a transaction. So, if you must do a transaction that updates the blockchain, you would have to shell out gas, and that gas costs ethers.

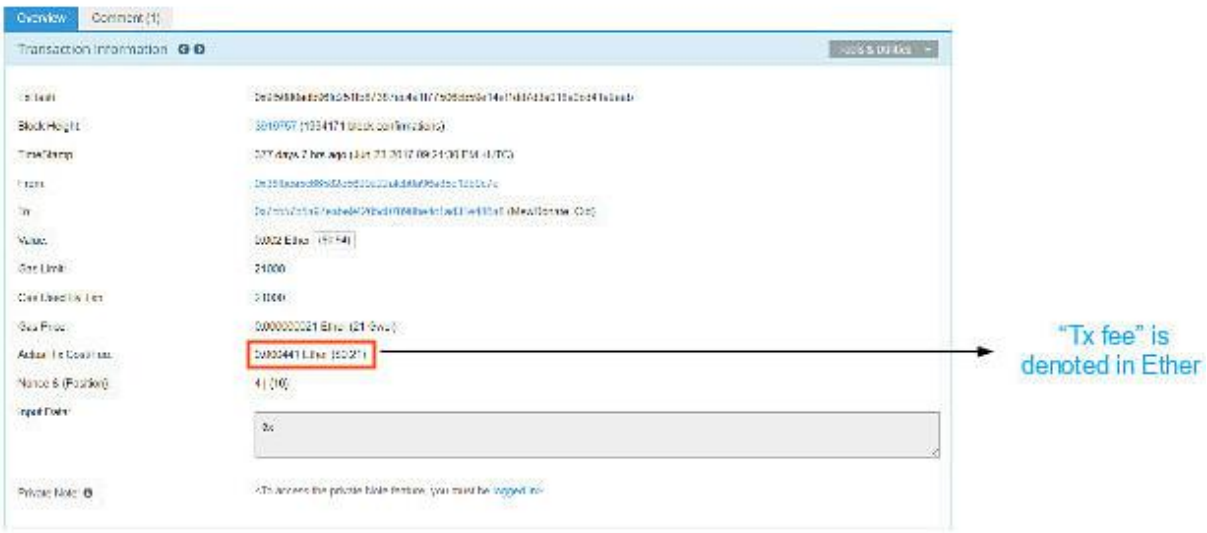
- In Ethereum, the transaction fees are calculated using a formula (see screenshot below). For every transaction, there is gas and its correlated gas price. The transaction fees equal the amount of gas required to execute a transaction multiplied by the gas price. "Gas limit" refers to the amount of gas used for the computation and the amount of ether a user is required to pay for the gas.





- Below is a screenshot from the Ethereum network showing the transaction cost. You can see for this particular transaction, the gas limit was 21,000, the gas used by the transaction was 21,000, and the gas price was 21 Gwei, which is the lowest denomination of ether. So, 21 Gwei * 21,000 gave the actual transaction fees: 0.000441 ethers, or about 21 cents as of today. As mentioned, the transaction fee goes to the miner, who has validated the transaction.

For example, below is a screenshot of Ether transaction, where the cost of transaction fee is shown



Note: "Tx Fee" is paid by users to miners

- To understand the gas limit and price, let's consider an example using a car. Suppose your vehicle has a mileage of 10 kilometers per liter and petrol costs $1 per liter. Under these parameters, driving a car for 50 kilometers would cost you five liters of petrol, which is worth $5. Similarly, to perform an operation or to run code on Ethereum, you need to obtain a certain amount of gas, like petrol, and the gas has a per-unit price, called gas price.

- If the user provides less than the amount of gas to run an operation, then the process will fail, and the user will be given the message "out of gas." And Gwei, as noted above, is the lowest denomination of ether used for measuring a unit of a gas price.

KEY TAKEAWAYS

- On the Ethereum blockchain, gas refers to the cost necessary to perform a transaction on the network.
- Gas prices are based on supply and demand for the network's validation requests.
- Transaction prices are based on the gas limit and gas price.

GAS - https://github.com/djrtwo/evm-opcode-gas-costs

Opcode - https://github.com/djrtwo/evm-opcode-gas-costs/blob/master/opcode-gas-costs_EIP-150_revision-1e18248_2017-04-12.csv

Decentralized Autonomous Organizations (DAOs)

A DAO is a digital organization that operates without hierarchical management; it works in a decentralized and democratic fashion. So basically, a DAO is an organization in which the decision-making is not in the hands of a centralized authority but preferably in the hands of certain designated authorities or a group or designated people as a part of an authority. It exists on a blockchain network, where it is governed by the protocols embedded in a smart contract, and thereby, DAOs rely on smart contracts for decision-making—or,

we can say, decentralized voting systems—within the organization. So, before any organizational decision can be made, it must go through the voting system, which runs on a decentralized application.

Here's how it works. People add funds through the DAO because the DAO requires funding in order to execute and make decisions. Based on that, each member is given a token that represents that person's percentage of shares in the DAO. Those tokens are used to vote in the DAO, and the proposal status is decided based on the maximum votes. Every decision within the organization must go through this voting process.

DAO vs Organization

DAO	A traditional organization
Fully democratized.	Usually hierarchical.
Voting required .	Voting may or may not require.
No trusted intermediary to count vote.	Outcome of voting must be handled manually.
Services offered are handled automatically.	Requires human handling, or centrally controlled automation.
All activity is transparent and fully public.	Activity is typically private, and limited to the public.

Real-World Applications of Ethereum

- **Voting Systems**
  As we've seen with DAO, voting systems are adopting Ethereum. The results of polls are publicly available, ensuring a transparent and fair democratic process by eliminating voting malpractices.
- **Banking Systems**
  Ethereum is getting adopted widely in banking systems because with Ethereum's decentralized system; it is challenging for hackers to gain unauthorized access. It also allows payments on an Ethereum-based network, so banks are also using Ethereum as a channel to make remittances and payments.
- **Shipping**
  Deploying Ethereum in shipping helps with the tracking of cargo and prevents goods from being misplaced or counterfeited. Ethereum provides the provenance and tracking framework for any asset required in a typical supply chain.
- **Agreements**
  With Ethereum smart contracts, agreements can be maintained and executed without any alteration. So in an industry that has fragmented participants, is subject to disputes, and requires digital contracts to be present, Ethereum can be used as a technology for developing smart contracts and for digitally recording the agreements and the transactions based on them.

Ethereum Accounts

An Ethereum account is similar to a bank account, but for ethers or ETH, where Ethereum can be held, transferred to other accounts, and can also be used to execute smart contracts. An Ethereum account is an entity that is composed of an Ethereum address along with a private key. The first 20 bytes of the SHA3 hashed public key is the Ethereum address.

Different Fields in Ethereum Accounts

1. **Nonce:** The nonce in an Ethereum account indicated the number of transactions that have been sent from that account. This ensures that each transaction is made only once by taking count every time it takes place.
2. **Ether Balance:** The balance of an Ethereum account denotes the amount of ether present in an ether repository of the current ether account.
3. **Contract Code:** This is non-mandatory to fill, in case it is present since not all accounts have a contract code. But note, that they cannot be altered once executed.
4. **Storage:** This field remains not filled unless mentioned.
5. **Code Hash:** The value of the code hash for Contract Accounts will be a hash that refers to the code present in that Ethereum account and since no code is associated with externally owned Ethereum accounts, therefore, code hash will be an empty string.

Types of Ethereum Accounts

Below are the two types of Ethereum Accounts:

1. Externally Owned Account: This is the most basic type of Ethereum account, it functions similarly to a bitcoin account. A private key controls the Ethereum address for EOAs. A person can open as many EOAs as they require. It is created whenever a wallet is created, and it is made with a private key that is required to access EOAs, check balances, send and receive transactions, and establish smart contracts.

Advantages:

1. Transactions from an external account to a contract account can trigger code that can execute many different actions, such as transferring tokens or even creating a new contract.
2. Externally Owned Accounts cannot list incoming transactions.

2. Contract-Based Account: Contract-based accounts can perform all of the functions of an externally owned account, but unlike EOAs, they are formed when a contract code is deployed, are governed by contract codes, and are accessed using a unique address. When one party accepts a contract, a unique account is formed which contains all of the charges associated with that contract. Each contract is granted a distinct serial number, which is referred to as a contract account.

Types of Contract Accounts

Below are the three types of contract accounts:

1. **Simple Account:** The account is created and owned by a single account holder.
2. **Multisig (multisignature) Account:** A Multisig Wallet contains several owner Accounts, one of which is also the creator Account.

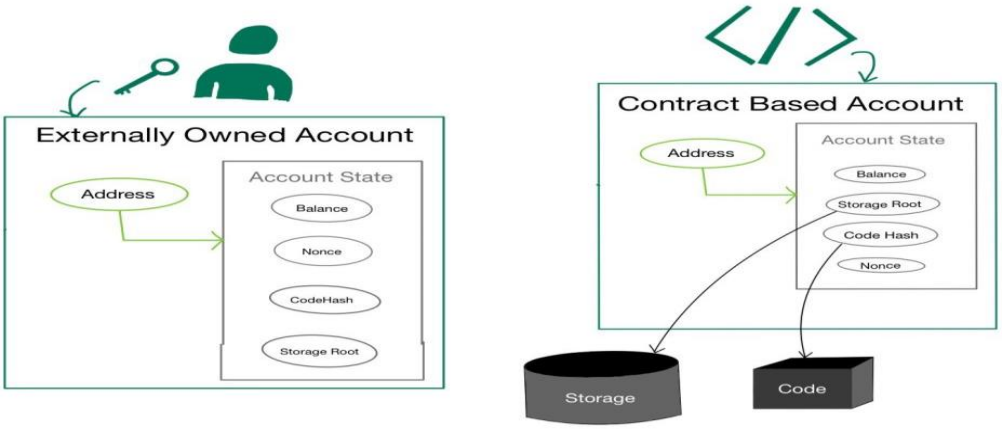
3. **Simplest Account:** A Multisig Wallet contains several owner Accounts, one of which is also the creator Account.

Advantages:

1. A contract account can list incoming transactions.
2. Contract accounts can be set up as Multisig Accounts.
3. A Multisig Account can be structured such that it has a daily limit that you specify, and only if the daily limit is exceeded will multiple signatures be required.

Disadvantages:

1. Creating contract accounts costs gas because they use the valuable computational and storage resource of the network.
2. Contract accounts can't initiate new transactions on their own. Instead, contract accounts can only fire transactions in response to other transactions they have received either from an externally owned account or from another contract account.



Externally Owned Accounts vs Contract Based Accounts

Below are the differences between Externally owned accounts and contract-based accounts.

S. No.	Externally Owned Accounts	Contract Accounts
1.	This account is controlled by humans.	This account is controlled Contract Code.
2.	The private key is needed to access EOAs.	No key is needed to access Contract Accounts.
3.	EASs are created automatically on creating a wallet.	CA require EOAs to be activated.
4.	EOAs do not have their own associated code.	CAs have their own associated code.
5.	No execution fee is associated with EOAs.	The execution fee is associated with CAs.
6.	Code hash is an empty string.	Code hash represents the code associated with the account.

Consensus Mechanisms Proof of Stake (PoS):

Proof-of-stake is a consensus mechanism used to decide the next global change in the network. It uses a staking mechanism where participants lock up some of their coins to get selected. Proof-of-stake in the blockchain is one of the controversial consensus protocols.

What is Proof-of-Stake (PoS) in blockchain?

Proof-of-stake overcomes the limitations of proof-of-work like excessive power and energy consumption. In the case of proof-of-work, all the miners have to do the heavy mathematical computation, and eventually, only one of them selects. Hence, this way results in massive wastage of computational and energy resources.

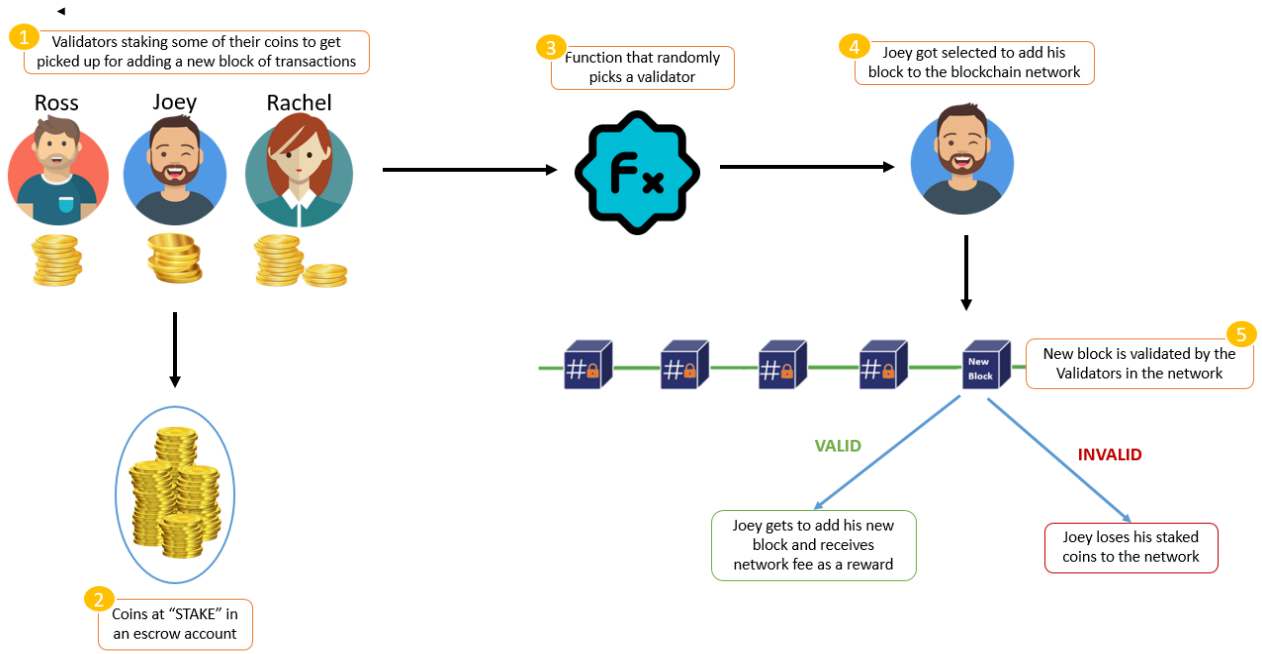
Therefore, unlike PoW, proof-of-stake (PoS) first stakes some coins. That means the participants (or **validators**) must give some of their coins as a **stake**. If the block they wish to forge in the network turns out to be fraudulent transactions, they'll lose that stake. Else, they'll get their staked coins.
Note: Proof-of-stake is a mechani*sm for the network participants who wish to add a new block of transactions. The purpose is to save the costs of mining a new block in the network from the beginning.*
In addition, in proof-of-stake, we have validators**. Validators** are basically the network participants who want to add their block of transactions. They must stake some of their coins in the escrow account to ensure their selection.
Hence, there are several validators selected. Based on random selection, one of them gets the chance to add his block. First, other validators check whether the chosen validator's block is valid by **attesting**. If marked as valid, he gets the reward. Else he loses his stake.

Let's understand the working of Proof-of-stake and the validation process thoroughly.

How does the Proof-of-Stake Algorithm work?

- Blocks in PoS, not mined but **forged** in PoS protocol.
- Already added network participants who wish to forge or add a new block of transactions can participate. They are also called **validators**.
- Validators have to give or initially lock up a portion of their earned coins as a **stake** in the escrow account for a time frame.
- One of the validators' blocks selected randomly based on their stakes.
- The selected block will be verified and **attested** by other validators.
    - Suppose a block attested by all the validators as **valid**. The block adds to the blockchain network, and its validator earns rewards.
    - Otherwise, if it's fraudulent or creates any discrepancy, it will be marked as **invalid**, and the participant will lose his staked coins.

Let's take an example of the above steps using a diagram.



Q. How Proof-of-stake comes with a number of improvements to the now-deprecated proof-of-work system:

- Better energy efficiency – there is no need to use lots of energy on proof-of-work computations
- Lower barriers to entry, reduced hardware requirements – there is no need for elite hardware to stand a chance of creating new blocks
- Reduced centralization risk – proof-of-stake should lead to more nodes securing the network

---

- Incentives - Because of the low energy requirement less ETH issue is required to incentivize participation
- Economic penalties for misbehavior make 51% style attacks exponentially more costly for an attacker compared to proof-of-work
- The community can resort to social recovery of an honest chain if a 51% attack were to overcome the crypto-economic defenses

PROS and CONS of PoS:

Pros	Cons
Staking makes it easier for individuals to participate in securing the network, promoting decentralization. validator node can be run on a normal laptop. Staking pools allow users to stake without having 32 ETH.	Proof-of-stake is younger and less battle-tested compared to proof-of-work
Staking is more decentralized. Economies of scale do not apply in the same way that they do for PoW mining.	Proof-of-stake is more complex to implement than proof-of- work
Proof-of-stake offers greater crypto-economic security than proof- of-work	Users need to run three pieces of software to participate in Ethereum's proof-of-stake.

Difference between Proof of Work (PoW) and Proof of Stake (PoS) in blockchain :

S.No.	Proof of Work (PoW)	Proof of Stake (PoS)
1.	The probability of mining a block is determined by how much computational work is done by miner.	The probability of validating a new block is determined by how large of a stake a person holds (how many coins they possess).
2.	A reward is given to first miner to solve cryptographic puzzle of each block.	The validator donot receive a block reward instead they collect network fee as their reward.
3.	To add each block to chain, miners must compete to solve difficult puzzles using their computer process power	There is no competition as block creator is .chosen by an algorithm based on user stake.
4.	Hackers would need to have 51% of computation power to add malicious block.	Hackers would need to own 51% of all cryptocurrency on network, which is practically impossible.
5.	Proof of work systems are less energy efficient and are less costly but more proven.	Proof of Stake systems are much more cost and energy efficient than POW systems but less proven.
6.	Specialized equipment to optimize processing power.	Standard server grade unit is more than enough.
7.	Initial investment to buy hardware.	Initial investment to buy stake and build reputation.
8.	Bitcoin is most well-known crypto with a Proof-of-Work consensus building algorithm which uses most well-known proof-of-work function is called SHA256.	Some of cryptocurrencies that use different variants of proof-of-stake consensus are: EOS (EOS), Tezos (XTZ), Cardano (ADA), Cosmos (ATOM), Lisk (LSK).

MetaMask:

- Meta Mask is Decentralized wallet founded by Aaron Davis in 2016.
- A New York based company named ConsenSys is the owner of MetaMask.
- MetaMask is a cryptocurrency wallet that enables users to store Ether and other ERC-20 tokens.
- MetaMask is a cryptocurrency wallet that enables users to access the Web 3 ecosystem of decentralized applications (dapps).
- MetaMask is a browser plugin that serves as an **Ethereum** wallet, and is installed like any other browser plugin. Once it's installed, it allows users to store Ether and other ERC-20 tokens, enabling them to transact with any Ethereum address.
- By connecting to MetaMask to Ethereum-based dapps, users can spend their coins in games, stake tokens in gambling applications, and trade them on decentralized exchanges (DEXs). It also provides users with an entry point into the emerging world of decentralized finance, or **DeFi**, providing a way to access DeFi apps such as Compound and PoolTogether.

How to install MetaMask on your computer?

You can set up MetaMask on your computer within a couple of minutes. Google Chrome, Firefox, Brave, Edge, or any Chromium-based browser will do. Let's take the example of the Chrome browser to help you understand better. To download and install MetaMask, follow the given steps:

1. Go to the MetaMask homepage, press 'Download', and click on 'Install MetaMask for Chrome'.

You will now be redirected to the Chrome web store.

1. Click on 'Add to Chrome'
2. Select 'Get Started' and then click on 'Create a wallet'
3. Then select any one of the options 'No Thanks/I Agree'
4. Create a strong password; it must be at least 8 characters long- PritiGolar123
5. Confirm your password and check the box to accept the Terms of Use
6. Hit the 'Create' button

You will be given a 12-word backup phrase. Copy it down somewhere safe. Anyone with access to this phrase can gain entry into your account. So, it is crucial to keep it very safe.

1. Now, click on 'Next'
2. Confirm your backup phrase by entering it in the correct order
3. Click on 'All Done'

You are now ready to buy, sell and swap tokens using your MetaMask crypto wallet.
For an android phone, you can download the MetaMask – Blockchain Wallet app from the Google Play Store. For an iPhone, you can download the same from the App Store.

How to link MetaMask to a crypto exchange?

You need to connect your crypto wallet to the exchange server to buy and sell tokens from a crypto exchange. Let's take the example of Uniswap. You can follow the steps given below:

1. Go to Uniswap.org and click on 'Use Uniswap'
2. Click on the 'Connect to a wallet' button
3. From the list, select a wallet of your choice (MetaMask in your case)
4. Enter your MetaMask password
5. Your wallet is now connected to Uniswap. You can now buy and sell tokens from this exchange.

How to buy and sell MetaMask tokens?

To buy tokens with MetaMask, you must deposit either Ether or any ERC-20 token. The steps are as follows:

- Go to the MetaMask extension on your browser and enter your password to unlock
- On the front page, click on the 'Buy' button and click on 'Directly Deposit Ether'
- Press the 'View Account' button located under it

You will see the wallet address and the QR code.

- Copy the wallet address and go to a crypto exchange of your choice to purchase tokens
- Click on the 'Send' option in the crypto exchange, enter your Wallet ID and other required information
- Hit 'Send'

The crypto exchange will take some time to initiate and complete your transaction. It will take place through the Wyre or Transak payment gateway. With the Wyre payment gateway, you can pay with your debit/credit card or a bank transfer. After the payment is complete, the amount of Ether you purchased will be deposited into your crypto wallet.

Question Bank-

Que.No	Question	CO Mapped	BTL Level
Que. 1)	What is ethereum? Explain how ethereum is different than Bitcoin.	[CO3]	[BTL4]
Que. 2)	Differentiate between ethereum and Bitcoin	[CO3]	[BTL4]
Que. 3)	Write an history of ethereum	[CO3]	[BTL6]
Que. 4)	Describe the Consensus Mechanisms Proof of Stake (PoS) in detail	[CO3]	[BTL 4]
Que. 5)	Distinguish between PoW & PoS	[CO3]	[BTL4]
Que. 6)	Create the blockdigram for working process of consensus mechanism PoS, explain with the help of example	[CO3]	[BTL6]
Que. 7)	Compare & contrast w.r.t prons & cons of PoS	[CO3]	[BTL5]
Que. 8)	Describe smart contract? Explain working of smart contract in ethereum blockchain	[CO3]	[BTL4]
Que. 9)	Explain the Features of Smart Contracts in ethereum blockchain	[CO3]	[BTL4]
Que. 10)	Write the application of smart contract, explain each of them in detail	[CO3]	[BTL6]
Que. 11)	Compare & contrast Smart Contracts Vs. Traditional Contract Systems	[CO3]	[BTL6]
Que. 12)	Explain advantages of smart contracts in ethereum blockchain	[CO3]	[BTL3]
Que. 13)	Illustrate ethereum accounts. Also explain Different Fields in Ethereum Accounts	[CO3]	[BTL3]
Que. 14)	Describe ethereum accounts. Classify the ethereum accounts and explain each in detail	[CO3]	[BTL4]
Que. 15)	Differentiate between externally owned accounts & contract based accounts	[CO3]	[BTL4]
Que. 16)	Draw a transaction structure of ethereum blockchain, explain each field present in the structure.	[CO3]	[BTL6]
Que. 17)	Describe dApp with example.	[CO3]	[BTL4]
Que. 18)	How Does a dApp Work? Explain with neat sketch.	[CO3]	[BTL2]
Que. 19)	Write the benefits and limitations of dApps.	[CO3]	[BTL6]
Que. 20)	Explain the concept of EVM with example.	[CO3]	[BTL3]
Que. 21)	What Is Gas (Ethereum)?	[CO3]	[BTL4]

Que. 22)	Write a short note on DAOs	[CO3]	[BTL6]
Que. 23)	Compare & contrast DAO Vs. Organization	[CO3]	[BTL6]
Que. 24)	Write the application of Ethereum	[CO3]	[BTL6]
Que. 25)	What is metamask? How to setup meta mask in computer.	[CO3]	[BTL4]
Que. 26)	How to link MetaMask to a crypto exchange?	[CO3]	[BTL2]
Que. 27)	How to buy and sell MetaMask tokens?	[CO3]	[BTL2]